Practice workbook · What algorithms are and how we write them down

What an algorithm is, the ways it can be expressed (flowchart, pseudocode, written description, program code) and which suits an audience, and what a trace table is and why it is used.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-12

1. Question 6 · part a · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part a only.

6 Programmers share algorithms with different people and write algorithms for different reasons.

(a) A programmer is showing a new algorithm to a group of non-technical managers.

State an appropriate method for writing the algorithm.

Justify your answer.

(2)

Method

Justification

(b) **Figure 1** shows an algorithm that displays a string based on the number input by the user.

```
1  SEND ("Enter a number: ") TO DISPLAY
2  RECEIVE inNum FROM (INTEGER) KEYBOARD
3  IF ((inNum = 1) OR (inNum = 2)) THEN
4      IF (inNum = 1) THEN
5          SEND ("First") TO DISPLAY
6      ELSE
7          IF (inNum = 2) THEN
8              SEND ("Second") TO DISPLAY
9          END IF
10     END IF
11 ELSE
12     SEND ("Invalid input") TO DISPLAY
13 END IF
```

Figure 1

Give **one** reason why the selection statement on line 7 is not required.

(1)

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1. Question 6 · part a · 2 marks · Working and answer

Working and answer

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2. Question 4 · part a · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a only.

4 Algorithms can be used to perform calculations and to process data.

(a) State what is meant by the term **algorithm**.

(1)

.....

.....

(b) **Figure 3** shows an algorithm.

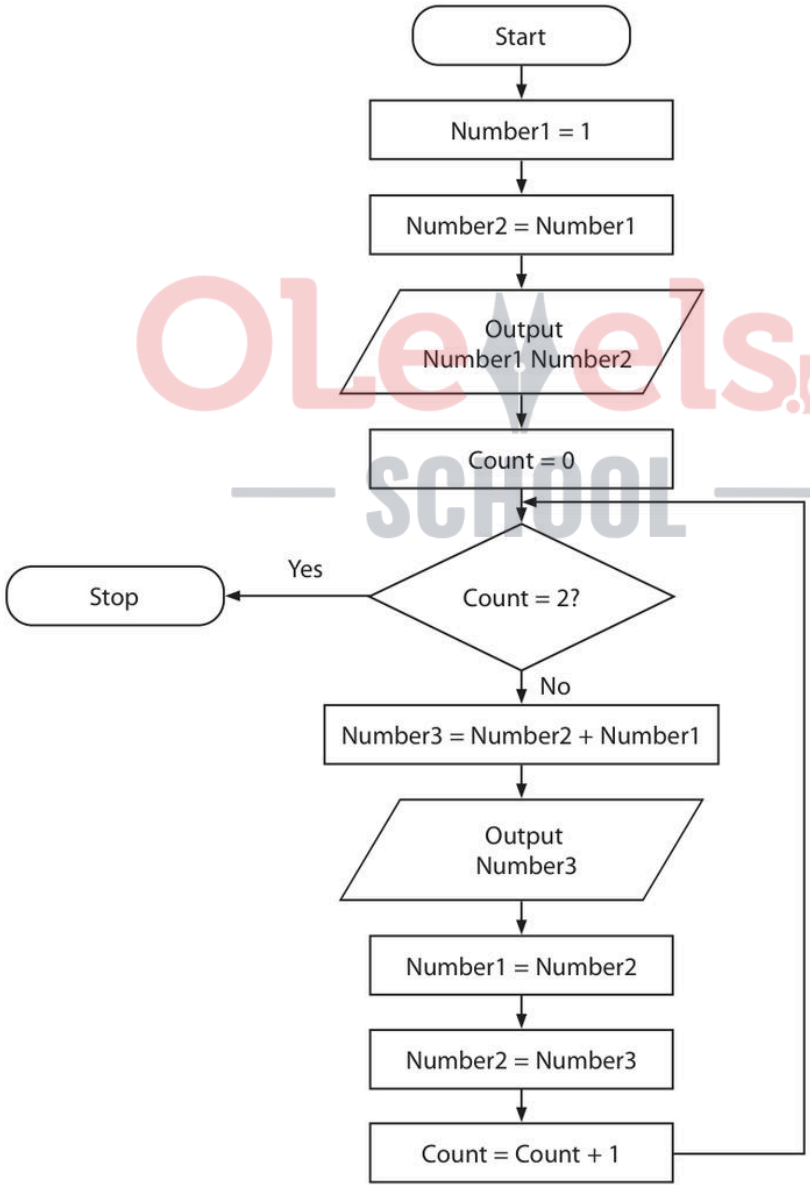


Figure 3

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P 6 9 2 9 7 A 0 1 2 2 0

2. Question 4 · part a · 1 mark · Working and answer

Working and answer

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3. Question 4 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

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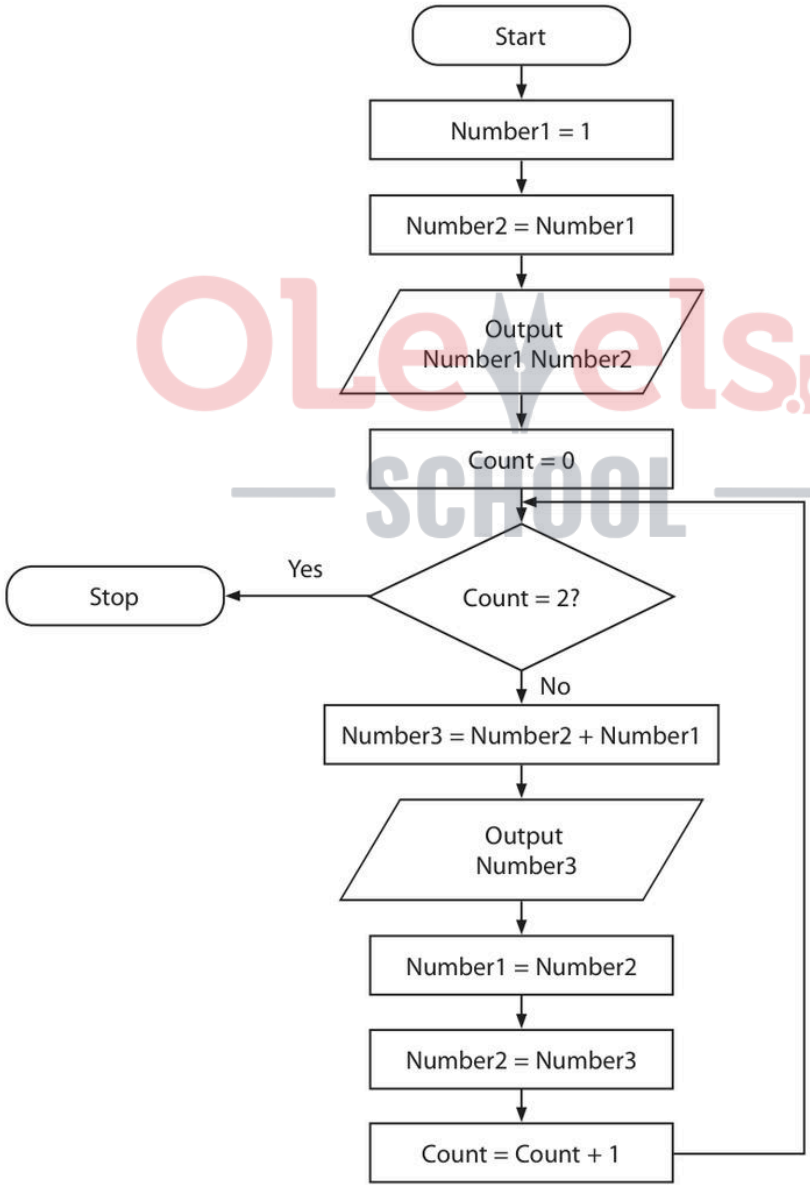


Figure 3

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DO NOT WRITE IN THIS AREA

P 6 9 2 9 7 A 0 1 2 2 0

3. Question 4 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

(i) Complete this trace table for the algorithm.

You may not need to use all the rows.

(5)

Number1	Number2	Number3	Count	Output	Count = 2?
1	1	–	0	1 1	False

(ii) The benefits of using a trace table include that they allow variable states, outputs and decisions to be recorded.

Give **one other** benefit of using a trace table to test an algorithm.

(1)

.....

.....

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P 6 9 2 9 7 A 0 1 3 2 0

13

Turn over

3. Question 4 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

(c) An algorithm is needed to count and display the number of vowels in a word.

The vowels are a, e, i, o, and u.

The completed algorithm must:

- count the number of vowels
- create a message as a single string (e.g. there are *number* vowels)
- print the message to the display.

Here is a partially completed algorithm written in pseudocode.

Complete the algorithm in the spaces provided.

(5)

```
SET word TO 'elephant'

FOR EACH letter FROM word DO

END FOREACH
```

(Total for Question 4 = 12 marks)

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3. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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4. Question 4 · part a.ii · 1 mark

2022 JUN · Paper 01

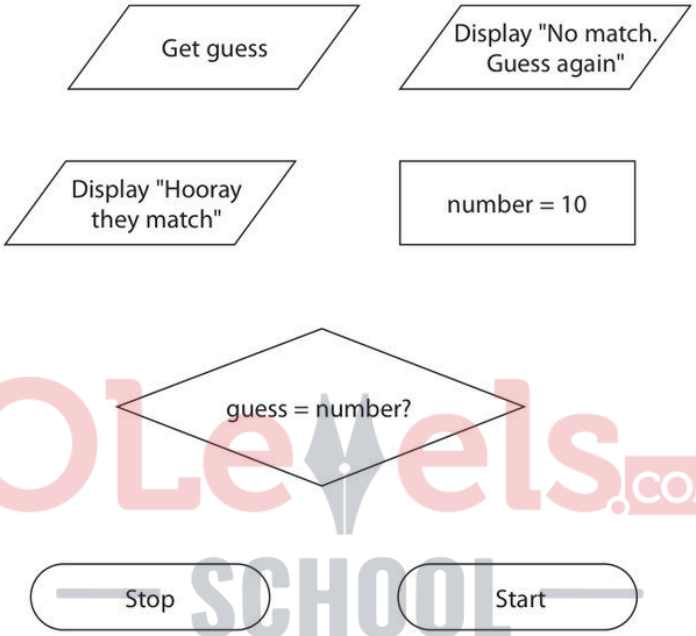
Further practice · Attempt part a.ii only.

4 Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.



Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(5)

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P 7 2 5 3 8 A 0 1 0 2 0

4. Question 4 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

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Draw your flowchart here.

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4. Question 4 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

	Step
A	Change the temperature to Celsius
B	Get the temperature in Fahrenheit
C	Show the temperature in Celsius
D	Set the temperature to 0

(i) Give the letter of the step that initialises a variable.

(1)

.....

(ii) Give the letter of the step that inputs a value.

(1)

.....

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P 7 2 5 3 8 A 0 1 2 2 0

4. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 3 · part a · 2 marks

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Further practice · Attempt part a only.

3 Algorithms can be represented in flowcharts, pseudocode or program code.

(a) Trace tables can be used with flowcharts or pseudocode.

Give **two** characteristics of a trace table.

(2)

1 .....

.....

2 .....

.....

(b) An algorithm has been written to validate numbers entered by the user.

Figure 2 shows the pseudocode for the algorithm.

```
1  RECEIVE num1 FROM (INTEGER) KEYBOARD
2  RECEIVE num2 FROM (INTEGER) KEYBOARD
3  IF (((num1 < 16) AND (num2 < 23)) OR (num2 = 13)) THEN
4      SEND "State 1" TO DISPLAY
5  ELSE
6      IF ((num2 > 12) AND (num1 > 20)) THEN
7          SEND "State 2" TO DISPLAY
8      ELSE
9          IF ((num1 = 88) OR NOT (num2 = 18)) THEN
10             SEND "State 3" TO DISPLAY
11         ELSE
12             SEND "State 4" TO DISPLAY
13         END IF
14     END IF
15 END IF
```

Figure 2

Complete the table to show the output for each set of inputs.

(3)

num1	num2	Output
88	18	
17	18	
12	19	

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5. Question 3 · part a · 2 marks · Working and answer

Working and answer

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6. Question 2 · part a.i · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a.i only.

2 Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

- ☒ **A** Algorithm
- ☒ **B** Computational thinking
- ☒ **C** Decomposition
- ☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

(ii) Give **one** example of a runtime error.

(1)

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6. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

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